

D7054E Project Assignment Proposal

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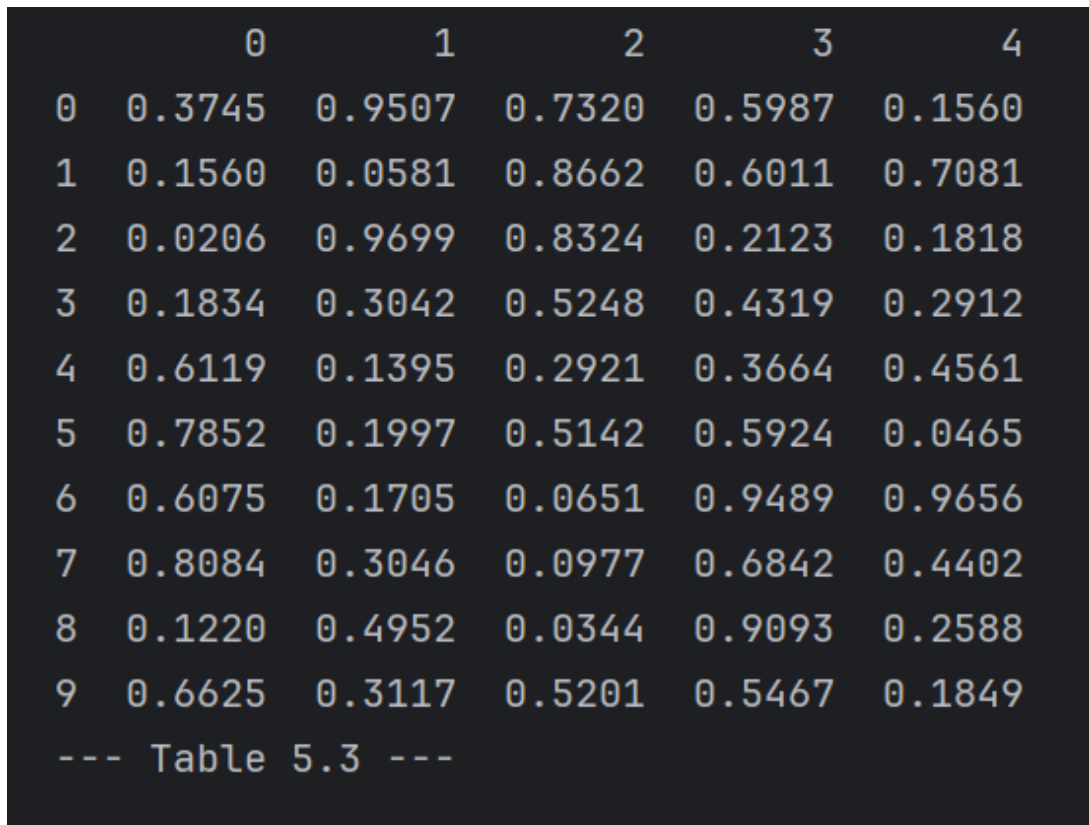
Group 16

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1 Continuous Distribution

1.1 Collect the Data

Use a random number generator to generate 50 values between zero and one (inclusive). Round the numbers to four decimal places.



	0	1	2	3	4
0	0.3745	0.9507	0.7320	0.5987	0.1560
1	0.1560	0.0581	0.8662	0.6011	0.7081
2	0.0206	0.9699	0.8324	0.2123	0.1818
3	0.1834	0.3042	0.5248	0.4319	0.2912
4	0.6119	0.1395	0.2921	0.3664	0.4561
5	0.7852	0.1997	0.5142	0.5924	0.0465
6	0.6075	0.1705	0.0651	0.9489	0.9656
7	0.8084	0.3046	0.0977	0.6842	0.4402
8	0.1220	0.4952	0.0344	0.9093	0.2588
9	0.6625	0.3117	0.5201	0.5467	0.1849

--- Table 5.3 ---

Figure 1: Table 5.3

1.2 Calculate the following:

- Mean (μ)
- Sample standard deviation (s)
- First quartile (Q_1)
- Third quartile (Q_3)
- Median

```
X_bar = 0.4459
s = 0.2889
First quartile = 0.1838
Third quartile = 0.6499
Median = 0.4360
```

Figure 2: Collect the Data calculation

1.3 Organize the Data

Construct a histogram of the empirical data using eight bars.

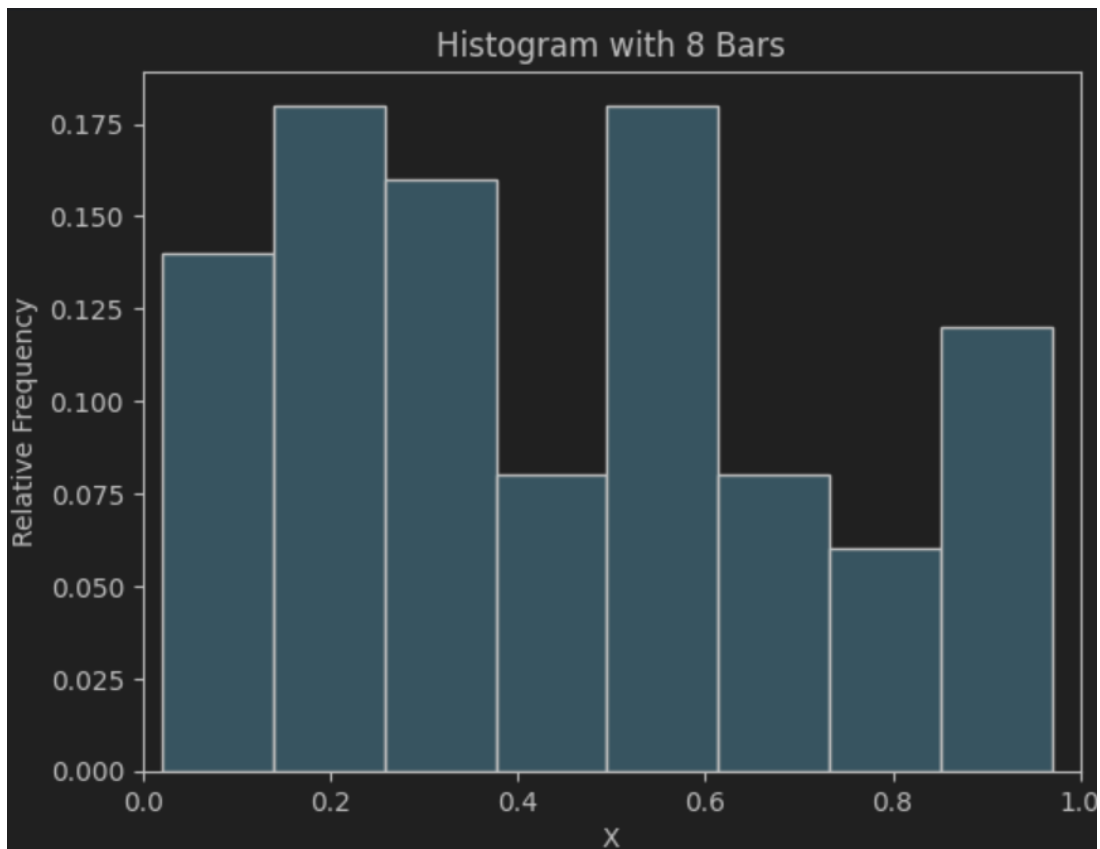


Figure 3: Figure 5.33

Construct a histogram of the empirical data using five bars.

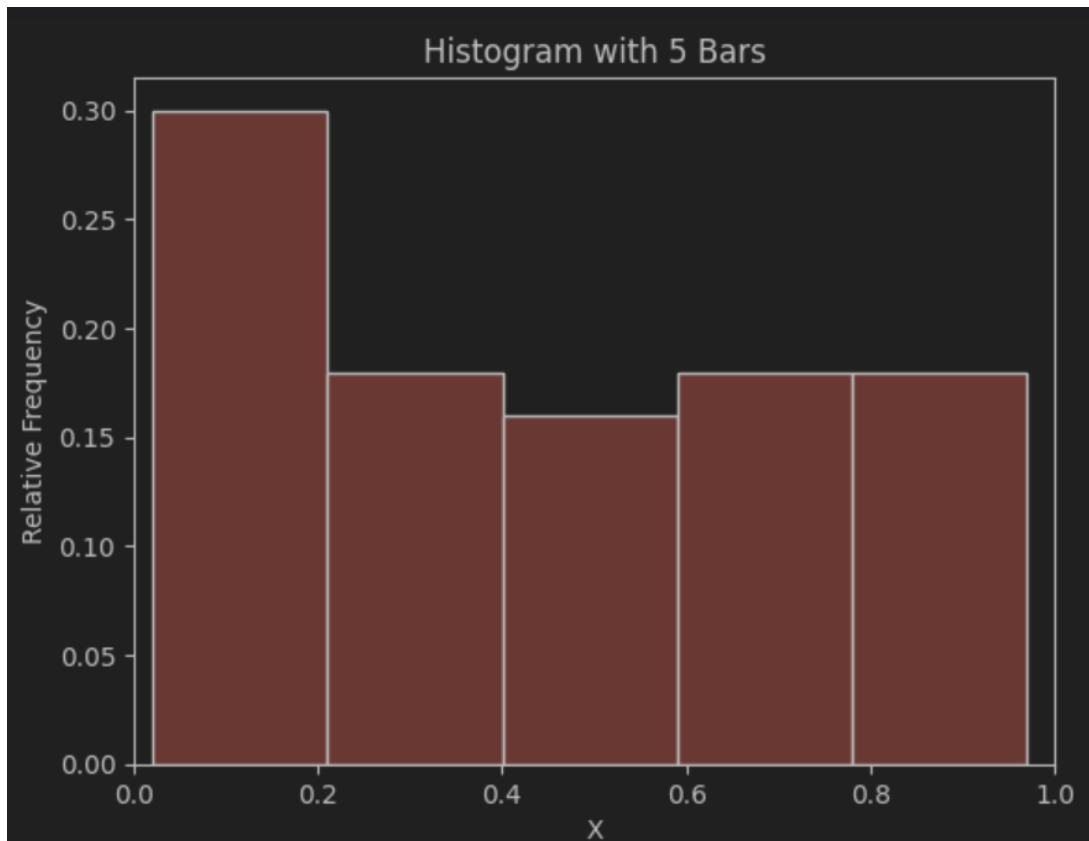


Figure 4: Figure 5.34

1.4 Describe the Data

Shape of Histogram (8 Bars)

The graph looks mostly flat but has several jagged peaks and dips across the top. If you draw a smooth curve through the bars, it goes slightly up and down but stays relatively straight across the whole range.

Shape of Histogram (5 Bars)

This graph shows hump at the very beginning on the left side. After that first high bar, the rest of the graph levels out and looks straight toward the end.

Effect of Changing Number of Bars

Changing the number of bars changes how "smooth" the shape looks because it groups the data differently. Having fewer bars makes the distribution look simpler and more blocky, while more bars show more of the small fluctuations and details in the data.

1.5 Theoretical Distribution

- In words, X = is a continuous random variable that follows a uniform distribution between zero and one.
- $\mu = 0.5$
- $\sigma = 0.2887$
- First quartile = 0.25
- Third quartile = 0.75

- **Median** = 0.5
- **Are the empirical values close to the theoretical values? Why or why not?**
Yes, the empirical values are reasonably close to the theoretical ones. For example, our sample standard deviation ($s = 0.2889$) is almost identical to the theoretical σ (0.2887). The other values, such as the mean (0.4459 vs. 0.5), differ slightly, but this is expected due to the randomness of a small sample size like 50.

1.6 Plot the Data

Construct a box plot of the data.

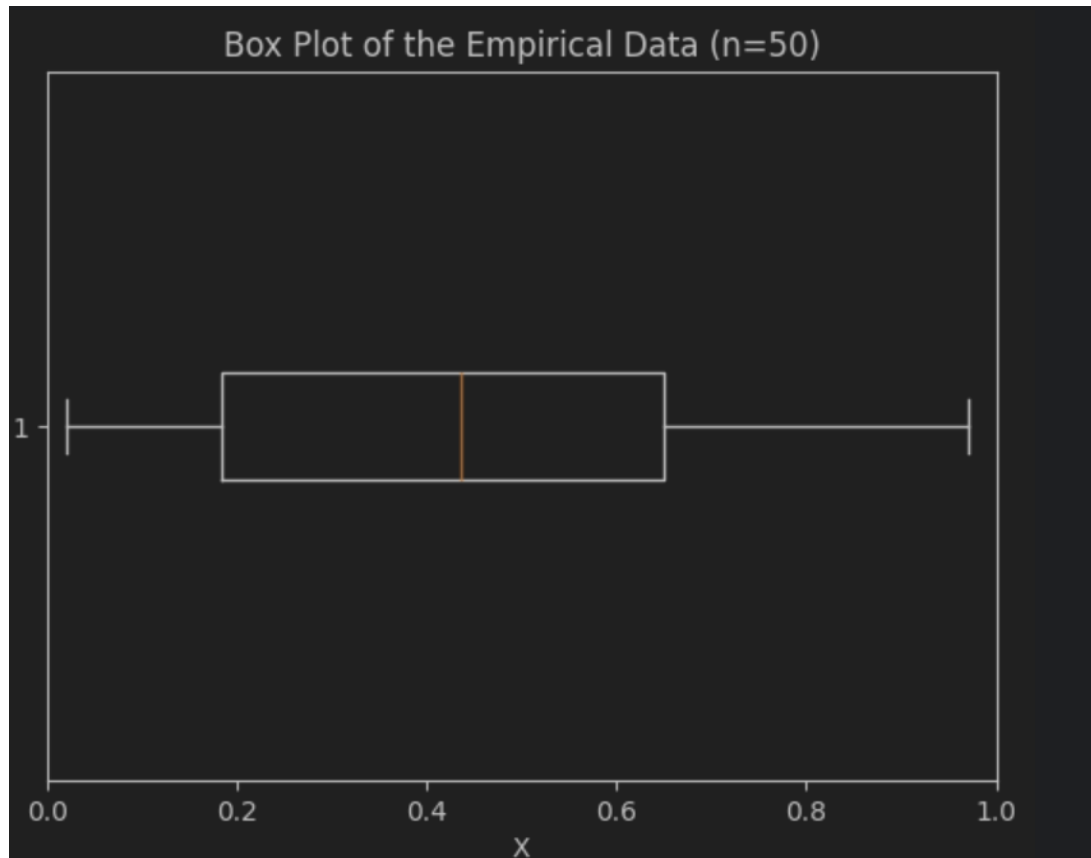


Figure 5: Box plot of the data

1.7 Outlier Analysis

- **$IQR = Q_3 - Q_1 = 0.6499 - 0.1838 = 0.4661$**
- **Lower Fence:** $0.1838 - 1.5(0.4661) = -0.5154$
- **Upper Fence:** $0.6499 + 1.5(0.4661) = 1.3491$

Do you notice any potential outliers?

No, there are no outliers. All data points lie between 0 and 1, which is well within the boundaries of -0.5154 and 1.3491 .

1.8 Compare the Data

- **Minimum value:** Our lowest value was about 0.0206, which is almost exactly what we expected since the distribution starts at zero.
- **First quartile:** The first quartile came out to 0.1838, which is a bit lower than the theoretical 0.25, but still shows the data is filling in the lower end.

- **Median:** The median was 0.4360, so it's quite close to the 0.5 we predicted, though slightly shifted.
- **Third quartile:** Our Q_3 was 0.6499, which is somewhat below the theoretical 0.75 mark.
- **Maximum value:** The maximum we got was 0.9699, which nearly reaches the theoretical upper limit of 1.
- **Width of IQR:** The IQR width was 0.4661, which is close to the theoretical width of 0.50.
- **Overall shape:** The graph doesn't look like a perfect rectangle because it's jagged, but it shows that the numbers are spread out fairly evenly across the whole range from 0 to 1, which is what we expect for this distribution.
- **How does the box plot fit the expected theoretical distribution?**
The box plot fits the theoretical distribution quite well because it shows the data is spread out widely across the whole range from 0 to 1 without any outliers. While the median isn't perfectly in the center of the box, this is because we are looking at a random sample of 50 values only.

1.9 Discussion Question

1.10 If 500 values were generated instead of 50, how would that affect:

- **The empirical statistics?**
The numbers would get much closer to the perfect theoretical values we expected. For example, the mean would likely be almost exactly 0.50 instead of being slightly off like it is now (0.4459).
- **The histogram shape?**
The histograms would lose their jagged, jumpy look and start to look like a solid and flat rectangle. Since there are 10 times more numbers.
- **The agreement with the theoretical distribution?**
The box plot looks way more symmetrical. The median line moves right into the center of the box, and the whiskers consistently reach the absolute edges of 0 and 1.

2 Normal Distribution (Lap Times)

2.1 Collect the Data

Use the data from Appendix C. Use a stratified sampling method by lap (races 1 to 20) and a random number generator to pick six lap times from each stratum. Record the lap times below for laps two to seven.

Lap2	Lap3	Lap4	Lap5	Lap6	Lap7
131	129	128	134	126	130
129	127	127	132	134	133
128	129	131	129	128	129
130	129	129	129	127	130
130	130	130	130	130	129
130	126	132	130	127	129

Figure 6: Table 6.1

Construct a histogram. Make five to six intervals. Sketch the graph using a ruler and pencil. Scale the axes.

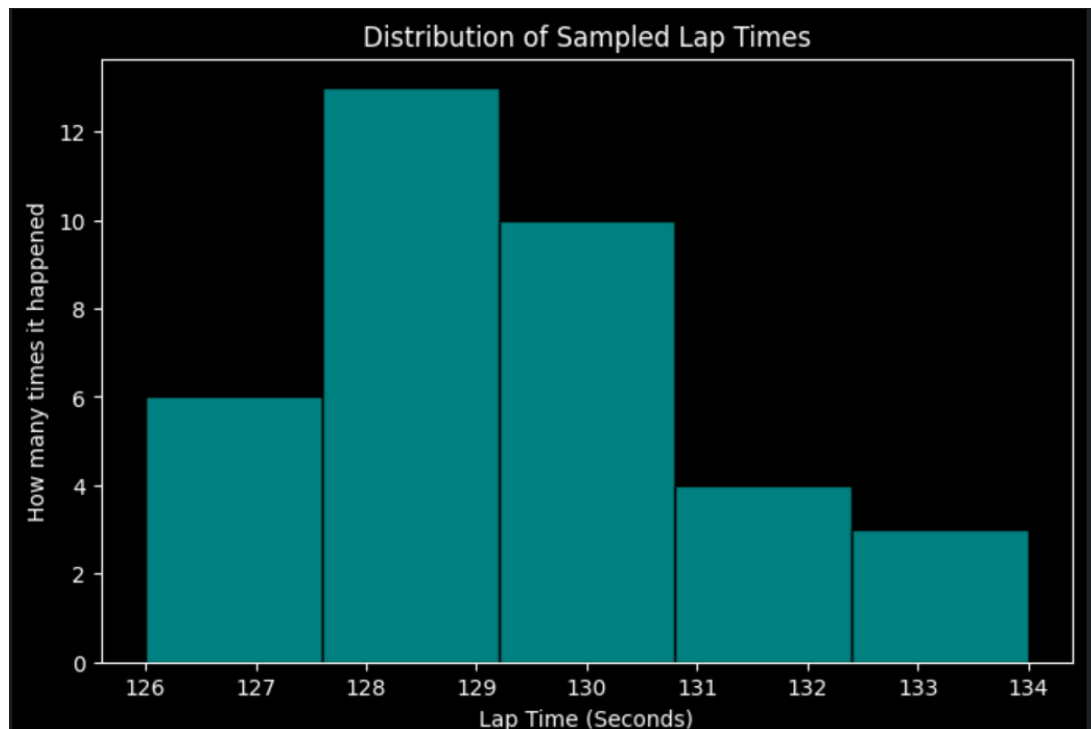


Figure 7: 6.10

2.2 Calculate the following

- $\bar{x}$
- s

```
Mean = 129.47222222222223
Std = 1.9344413901856763
```

Figure 8: Result

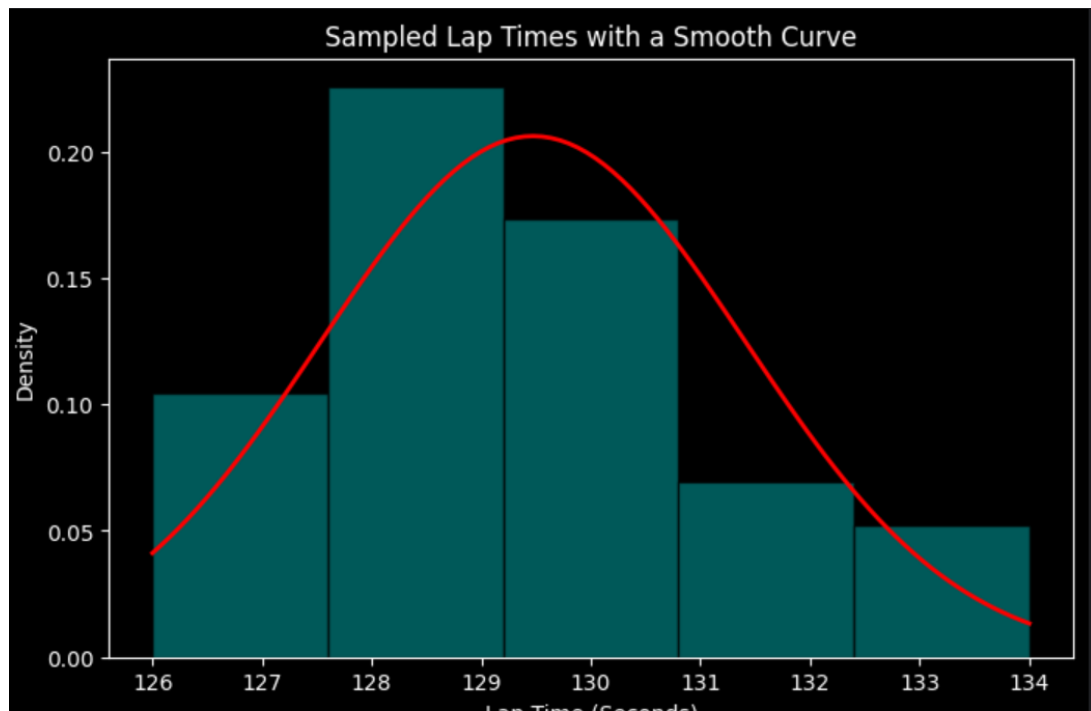


Figure 9: Smooth Curve

The curve peaks in the middle lap times and gradually slopes down as the times get higher or lower on both sides.

2.3 Analyze the Distribution

Using your sample mean, sample standard deviation, and histogram to help, what is the approximate theoretical distribution of the data?

$$X \sim N(\mu = 129.83, \sigma = 1.89)$$

How does the histogram help you arrive at the approximate distribution?

The histogram shows that most of the lap times are clustered in the middle, around 130 seconds, and they decrease as we move to the edges.

2.4 Describe the Data

Use the data you collected to complete the following statements.

- The IQR goes from - to -
129 to 131
- IQR =
2
- The 15th percentile is
128
- The 85th percentile is
132
- The median is
130
- The empirical probability that a randomly chosen lap time is more than 130 seconds is
0.2778
- Explain the meaning of the 85th percentile of this data.
85% of the sampled lap times were 132 seconds or less, while only 15% of the laps were slower than 132 seconds.

2.5 Theoretical Distribution

Using the theoretical distribution, complete the following statements.

You should use a normal approximation based on your sample data.

- The IQR goes from - to -.

128.56 to 131.11

- IQR =

2.55

- The 15th percentile is

127.87

- The 85th percentile is

131.79

- The median is

129.83

- The probability that a randomly chosen lap time is more than 130 seconds is

0.4641

- Explain the meaning of the 85th percentile of this distribution.

85% of all possible lap times are expected to fall at or below 131.79 seconds.

2.6 Discussion Questions

Do the data from the section titled Collect the Data give a close approximation to the theoretical distribution in the section titled Analyze the Distribution? In complete sentences and comparing the result in the sections titled Describe the Data and Theoretical Distribution, explain why or why not.

Yes, our data is a close match to the theoretical distribution. The sample median and percentiles (130, 128, and 132) are nearly identical to the theoretical targets of 129.8, 127.8, and 131.7. The only real difference is in the probabilities, which is normal since a small sample of 36 laps can't be as perfect as a mathematical curve.

3 Central Limit Theorem (Cookie Recipes)

3.1 Given

X = length of time (in days) that a cookie recipe lasted at the Olmstead Homestead. (Assume that each of the different recipes makes the same quantity of cookies.)

Calculate the following:

$$\mu_x = 3.58$$

$$\sigma_x = 2.16$$

3.2 Collect the Data

Use a random number generator to randomly select four samples of size $n = 5$ from the given population. Record your samples in Table 7.5 and then, for each sample, calculate the mean to the nearest tenth. Record them in the spaces provided. Record the sample means for the rest of the class.

```

Population Mean: 3.567
Population Std Dev: 2.148
-----

Samples for n=5:
Sample 1: [4 6 5 4 5] -> Mean = 4.8
Sample 2: [2 1 4 2 5] -> Mean = 2.8
Sample 3: [2 5 6 6 5] -> Mean = 4.8
Sample 4: [2 2 2 4 1] -> Mean = 2.2
Mean of means: 3.65
Std dev of means: 1.35

```

Figure 10: N=5 Results

Repeat with $n = 10$

Again, use a random number generator to randomly select four samples from the population. This time, make the samples of size $n = 10$. Record the samples in Table 7.6. As before, for each sample, calculate the mean to the nearest tenth. Record them in the spaces provided. Record the sample means for the rest of the class.

```

Samples for n=10:
Sample 1: [2 1 1 2 6 2 2 5 5 2] -> Mean = 2.8
Sample 2: [5 3 5 3 1 1 6 6 2 5] -> Mean = 3.7
Sample 3: [6 3 2 6 6 2 1 1 6 2] -> Mean = 3.5
Sample 4: [6 2 1 5 6 4 6 6 2 5] -> Mean = 4.3
Mean of means: 3.58
Std dev of means : 0.62

```

Figure 11: N=10 Results

Histogram for the Original Population

For the original population, construct a histogram. Make intervals with a bar width of one day. Sketch the graph using a ruler and pencil. Scale the axes.

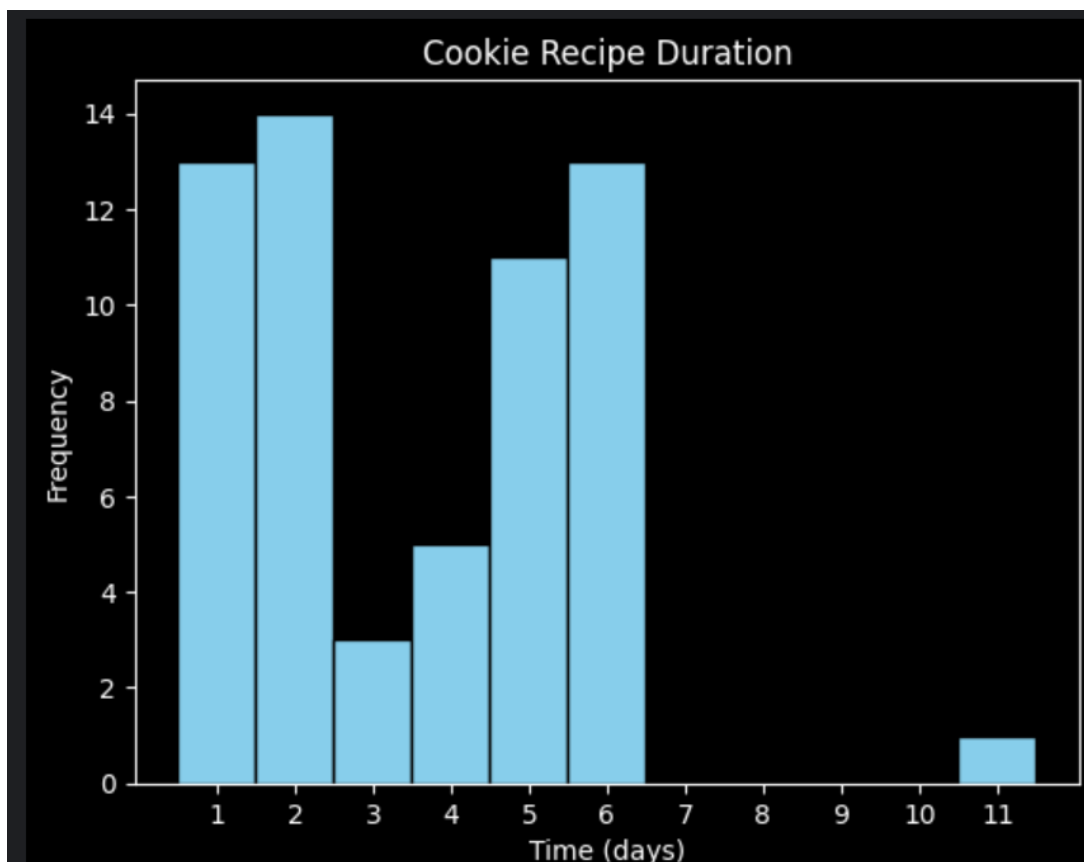


Figure 12:

Draw a smooth curve through the tops of the bars of the histogram. Use one to two complete sentences to describe the general shape of the curve.

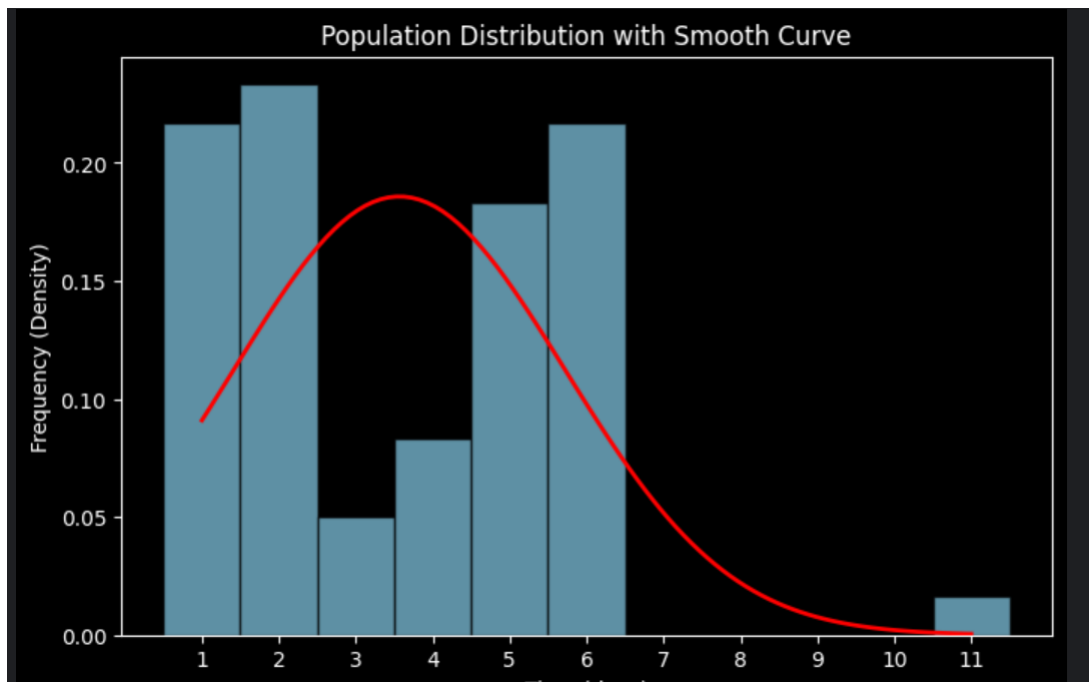


Figure 13:

Repeat the Procedure for $n = 5$

For the sample of $n = 5$ days averaged together, construct a histogram of the averages (your means together with the means of the other groups). Make intervals with bar widths of $\frac{1}{2}$ a day. Sketch the graph using a ruler and pencil. Scale the axes.

Draw a smooth curve through the tops of the bars of the histogram. Use one to two complete sentences to describe the general shape of the curve.

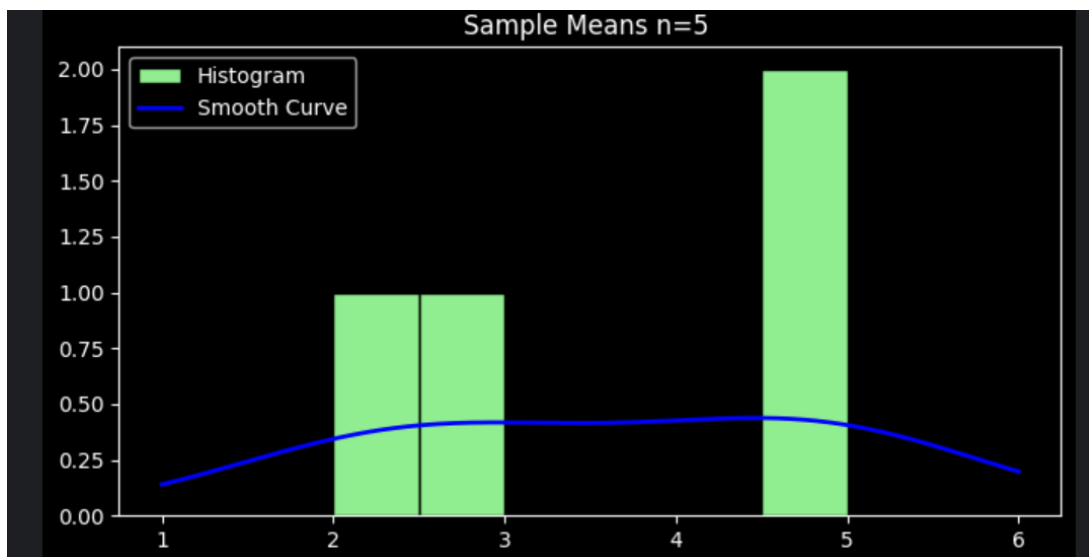


Figure 14:

The curve is relatively wide and flat, indicating a high spread among the sample means due to the small sample size. It shows that with only 5 recipes per sample, our results vary significantly from one group to another.

Repeat the Procedure for $n = 10$

For the sample of $n = 10$ days averaged together, construct a histogram of the averages (your means together with the means of the other groups). Make intervals with bar widths of $\frac{1}{2}$ a day. Sketch the graph using a ruler and pencil.

Scale the axes.

Draw a smooth curve through the tops of the bars of the histogram. Use one to two complete sentences to describe the general shape of the curve.

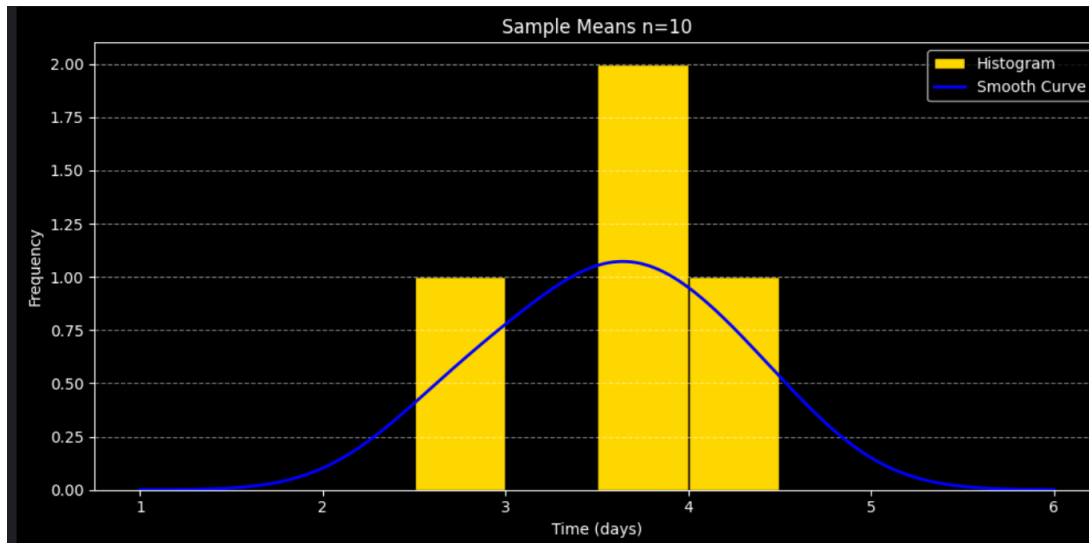


Figure 15:

The curve is noticeably narrower and taller, which means the sample means are clustering much closer to the actual population average. This demonstrates that increasing the sample size to 10 significantly reduces error and makes our estimation more precise.

Discussion Questions

Compare the three histograms you have made, the one for the population and the two for the sample means. In three to five sentences, describe the similarities and differences.

The population histogram is highly skewed and spread out, while the sample mean histograms become increasingly symmetric and narrow as n increases from 5 to 10.

State the theoretical (according to the CLT) distributions for the sample means.

$$n = 5 : \bar{x} \sim N(3.57, 0.96)$$

$$n = 10 : \bar{x} \sim N(3.57, 0.68)$$

Are the sample means for $n = 5$ and $n = 10$ “close” to the theoretical mean, μ_x ? Explain why or why not.

Yes, the sample means are very close to the theoretical mean (3.57) because the mean of a sampling distribution is always an unbiased estimator of the population mean.

Which of the two distributions of sample means has the smaller standard deviation? Why?

The $n = 10$ distribution has the smaller standard deviation because larger samples reduce the impact of individual outliers and lead to more consistent averages.

As n changed, why did the shape of the distribution of the data change? Use one to two complete sentences to explain what happened.

As n changed, the shape became more normal due to the Central Limit Theorem. This happens because the sum of random variables tends toward a normal distribution as more variables are added.

4 Explore the Melbourne Real-Estate Dataset

Perform any three data analysis tasks other than simple mean and standard deviations.

For example, use the dataset to see if there is any correlation between distance from CBD and prices of properties. Can you find/analyze something about house sizes and prices. How does it look for different suburbs in terms of number of properties, which have the most homes and least homes. Which councils/suburbs have homes built in which years? How does it look for different sellers when selling homes? How are the different regions in terms of price point and house sizes? Think creatively!

4.1 Distance vs. Price

We investigated whether being closer to the Central Business District (CBD) is still the primary driver of high property prices. Our analysis revealed a negative correlation of -0.15 between distance and price.

- While there is a general trend of prices dropping as you move away from the city center, the relationship is weak at -0.15
- The Human Reason suggests that Melbourne's market is "multi-polar." High-end suburbs, like those in the Bayside or Eastern regions, keep their high values no matter how far they are from the CBD. This is because of things like schools, lifestyle factors, and local amenities.

4.2 Market Hotspots: Where is the Most Activity?

Identifying where the most homes are being listed helps pinpoint "turnover hubs"—areas where the market is most liquid.

- Reservoir stands out as the most active suburb in the dataset with 359 listings.
- Other high-activity areas include Richmond (253), Bentleigh East (249), Preston (239), and Brunswick (220).
- These suburbs are the "heart" of the mid-market. Their high number of listings often means that the area is going through urban renewal or subdivision, which makes them prime targets for both first-time home buyers.

4.3 How Councils Have Changed Over Time?

Melbourne's chronological expansion can be mapped by examining the Median YearBuilt across various Council Areas.

- The median build years in councils like Melbourne and Yarra are significantly older.
- Outer councils show significantly more recent median build years, identifying them as newer residential developments and urban growth boundaries.
- This historical data is crucial because older homes in inner councils often command a "heritage premium," even if the land size is smaller than newer builds in the outskirts.

4.4 Results Screenshots

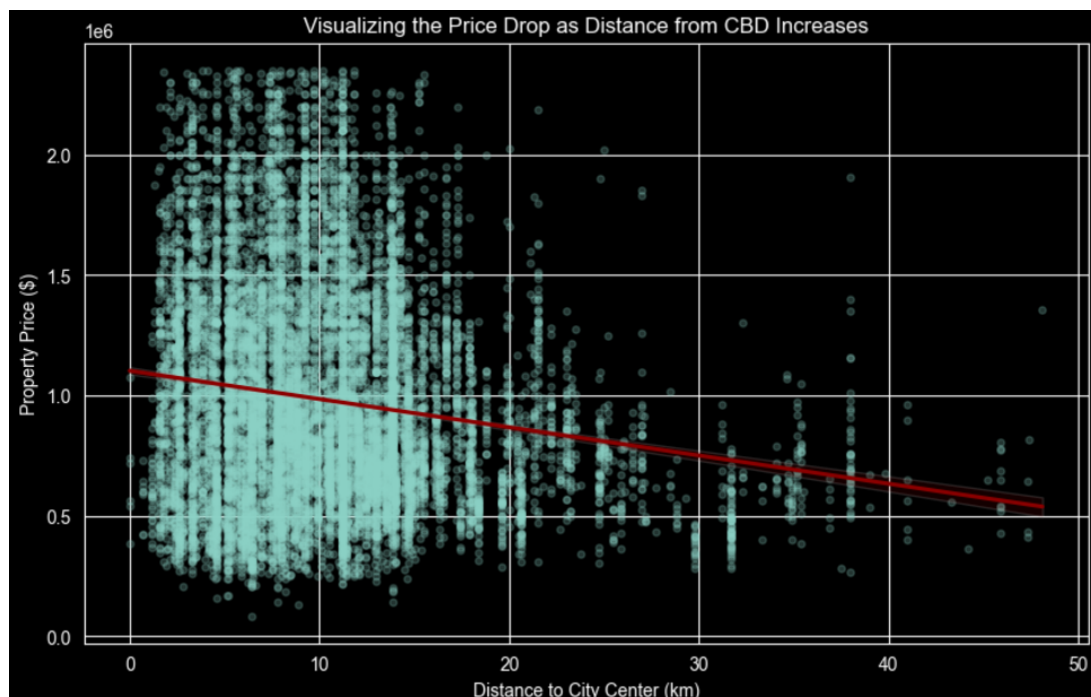


Figure 16: Visualizing the Price Drop as Distance from CBD Increases

Suburb	
Reservoir	359
Richmond	253
Bentleigh East	249
Preston	239
Brunswick	220

Figure 17: Top 5 Busiest Suburbs by Listing Volume

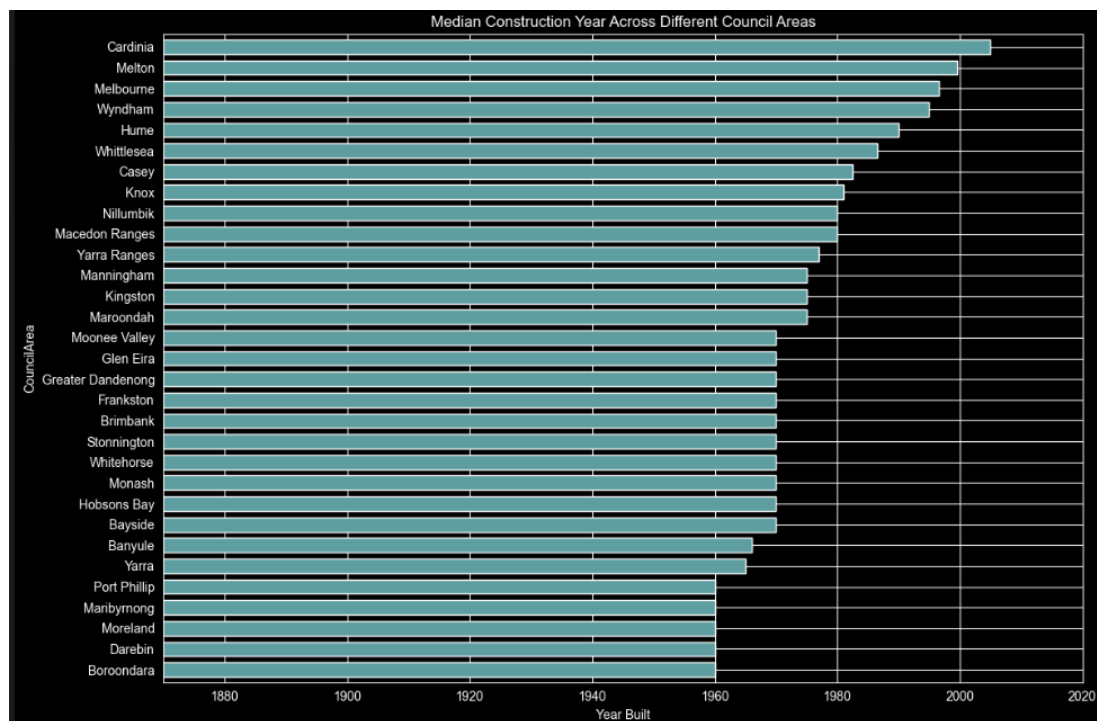


Figure 18: Median Construction Year Across Different Council Areas

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Distance vs Price correlation: -0.15 (negative means prices usually drop as distance increases).
Most listed suburb in this dataset: Reservoir (359 listings).
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Figure 19: Final Summary Output